



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

UNIT TEST

30-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2023

Corporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

☎ +91-744-2757575 ✉ info@allen.ac.in 🌐 www.allen.ac.in

SECTION-I : (Maximum Marks: 80)

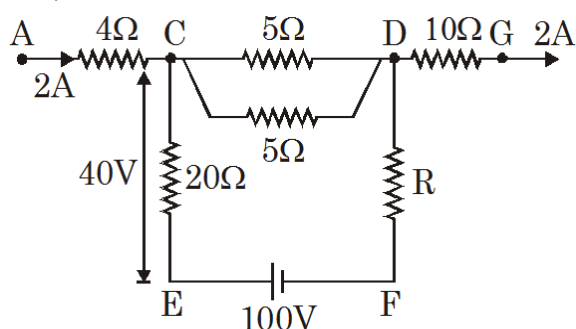
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

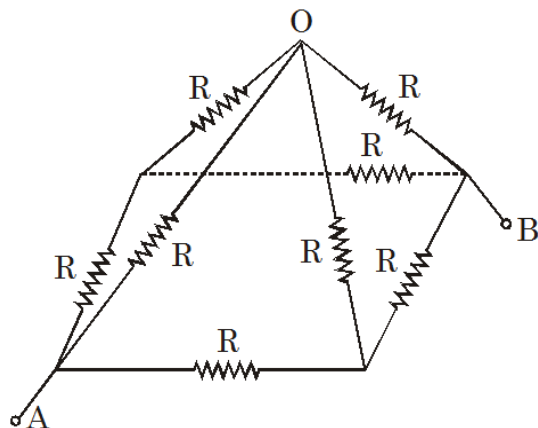
Negative Marks : -1 If wrong option is selected.

1. In the given circuit, find the value of resistance 'R' :



- (A) 20Ω (B) 25Ω
(C) 15Ω (D) 10Ω

2. In the given figure, $R = 3\Omega$. Determine equivalent resistance between points A and B.

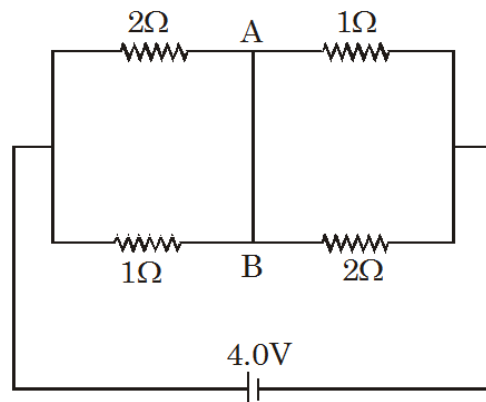


- (A) 3 Ω (B) 2 Ω
(C) 1.5 Ω (D) 6 Ω

3. A piece of conducting wire of resistance R is cut into 2n equal parts. Half the parts are connected in series to form a bundle and remaining half in parallel to form another bundle. These bundles are then connected to give the maximum resistance. The resistance of the combination is

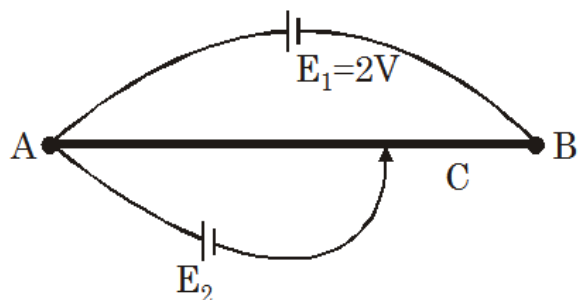
- (A) $\frac{R}{2} \left(1 + \frac{1}{n^2} \right)$
(B) $\frac{R}{2} (1 + n^2)$
(C) $\frac{R}{2(1 + n^2)}$
(D) $R \left(n + \frac{1}{n} \right)$

4. In figure, points A and B are connected by a perfectly conducting wire. Calculate the current through AB



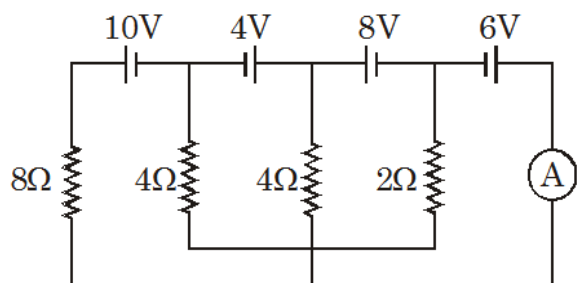
- (A) 2A
(B) 1 A
(C) 1.5 A
(D) 2.5 A

5. AB is a potentiometer wire of length 100 cm. When a cell E_1 is connected across AC, where $AC = 75$ cm, no current flows from E_2 . The internal resistance of the cell E_1 is negligible.



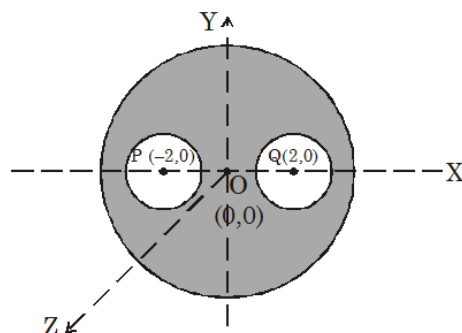
Potential gradient along AB and value of E_2 respectively is.

- (A) 0.01 V cm^{-1} , 1.2 volt
 (B) 0.03 V cm^{-1} , 1.3 volt
 (C) 0.04 V m^{-1} , 1.0 volt
 (D) 0.02 V cm^{-1} , 1.5 volt
6. Find the reading of the ideal ammeter connected in the given circuit. Assume that the cells have negligible internal resistance.

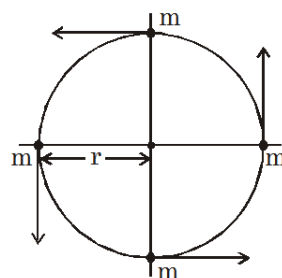


- (A) 2.5 A
 (B) 5 A
 (C) 2 A
 (D) none of these

7. A solid sphere of uniform density & having total mass (M) now two spherical cavity is removed as shown in figure. Radius of sphere is 4 meter. Radius of each spherical cavity is 1 meter. Value of gravitation field at centre of sphere will be :-

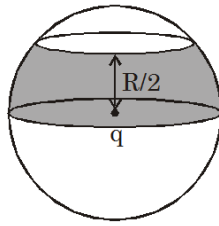


- (A) $\frac{31GM}{1024}$ (B) $\frac{31GM}{256}$
 (C) $\frac{GM}{1024}$ (D) Zero
8. Four similar masses orbiting around a circle of radius 'r'. Due to their mutual gravitation attraction force velocity of each particle will be :-



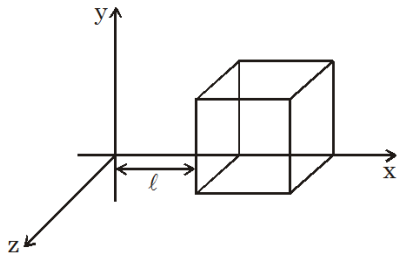
- (A) $\sqrt{\frac{Gm}{r} \left(\frac{1 + 2\sqrt{2}}{4} \right)}$
 (B) $\sqrt{\frac{Gm}{r} (1 + 2\sqrt{2})}$
 (C) $\sqrt{\frac{Gm}{r}}$
 (D) $\sqrt{\frac{Gm}{r} \left(\frac{1 + 2\sqrt{2}}{2} \right)}$

9. Find out flux passing through shaded surface of the sphere when a point charge 'q' is placed at centre of sphere. [Radius of sphere is R]



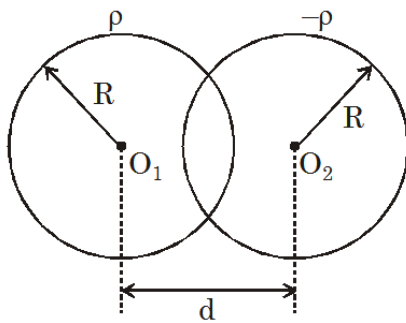
- (A) $\frac{q}{\epsilon_0}$ (B) $\frac{q}{4\epsilon_0}$ (C) $\frac{q}{2\epsilon_0}$ (D) Zero.

10. A cube of side ' ℓ ' placed as shown in figure $E_x = \beta x^{\frac{1}{2}}$. Flux passing through cube will be:-



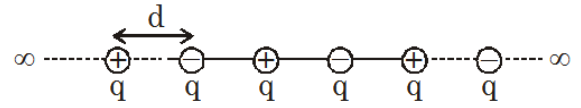
- (A) $2\sqrt{2}\beta\ell^{\frac{5}{2}}$ (B) $(\sqrt{2} - 1)\beta\ell^{\frac{5}{2}}$
(C) $-\beta\ell^{\frac{5}{2}}$ (D) Zero

11. Two solid spheres having volume charge density ρ and $-\rho$ respectively as shown in figure. Find out the potential difference (ΔV) between centre of both sphere for $d = R$



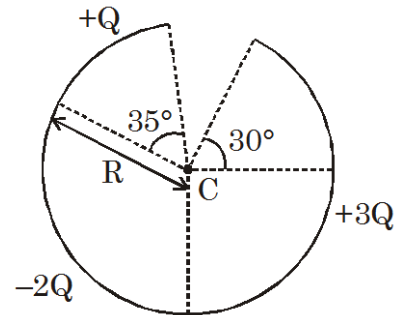
- (A) $\frac{\rho}{3\epsilon_0}R^2$ (B) $\frac{\rho}{\epsilon_0}R^2$ (C) 0 (D) $\frac{2\rho}{\epsilon_0}R^2$

12. There is a infinite chain of charge q & -q. Distance between the two neighbouring charge is equal to 'd'. The interaction energy of one of charge with all other charge is :



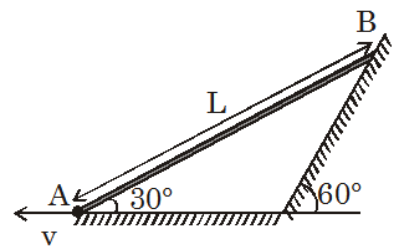
- (A) $\frac{-2q^2}{4\pi\epsilon_0 d}$ (B) $\frac{-2q^2 \ln 2}{4\pi\epsilon_0 d}$
(C) $\frac{2q^2 \ln 2}{4\pi\epsilon_0 d}$ (D) $\frac{q^2}{\pi\epsilon_0 d}$

13. Three arc are joined together as shown in figure having charges Q, -2Q, 3Q respectively. Find out the electric potential at point C.



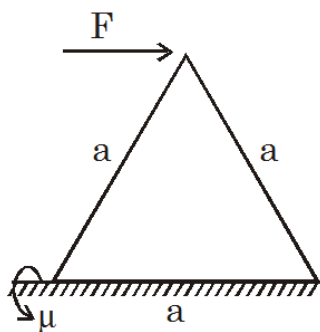
- (A) $\frac{Q}{2\pi\epsilon_0 R}$ (B) $\frac{5Q}{12\pi\epsilon_0 R}$
(C) $\frac{3Q}{32\pi\epsilon_0 R}$ (D) $\frac{3Q}{8\pi\epsilon_0 R}$

14. Instantaneous speed on end A of rod is 'v'. Angular velocity of rod of length 'L' will be.

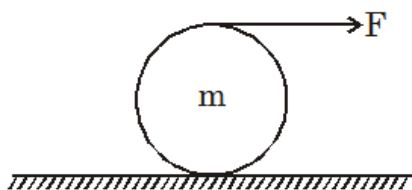


- (A) $\frac{v}{2L}$ (B) $\frac{v\sqrt{3}}{2L}$ (C) $\frac{v}{L}$ (D) $\frac{4v}{2L}$

15. Equilateral prism of mass m is rest on rough horizontal floor. If coefficient of friction is sufficiently high such that prism does not slide before topple. Minimum force required to topple prism is (coefficient of friction = μ)



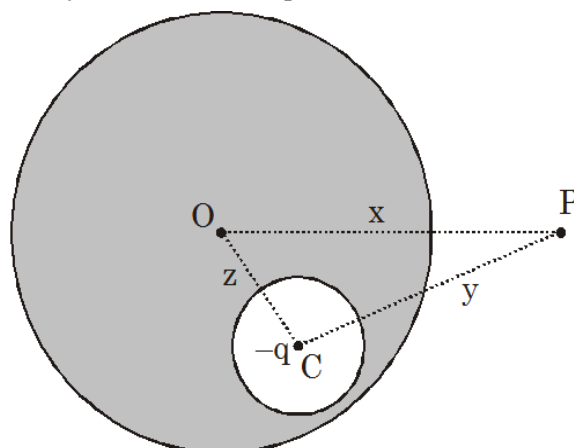
- (A) $\frac{mg}{\sqrt{3}}$ (B) $\frac{\mu mg}{\sqrt{3}}$
 (C) $\frac{mg}{4}$ (D) $\frac{\mu mg}{4}$
16. A force 'F' act tangentially at highest point of a disc of mass 'm' & radius 'R'. If disc is rolling without sliding. Acceleration of centre of disc will be



- (A) $\frac{2F}{3m}$ (B) 0 (C) $\frac{10F}{7m}$ (D) $\frac{4F}{3m}$
17. Time period of a simple pendulum whose length is $\frac{R}{2}$, where R is radius of the earth oscillating near earth's surface will be -

- (A) $2\pi\sqrt{\frac{R}{2g}}$ (B) $2\pi\sqrt{\frac{R}{g}}$
 (C) $2\pi\sqrt{\frac{R}{3g}}$ (D) $2\pi\sqrt{\frac{3R}{g}}$

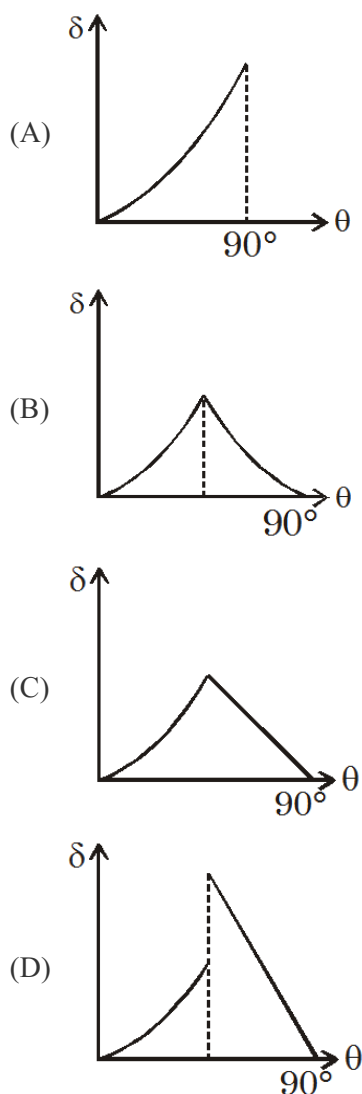
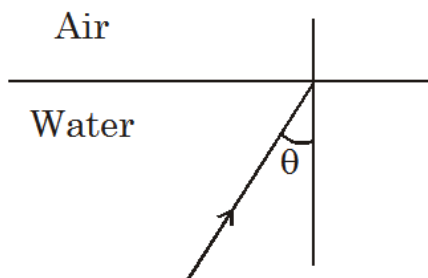
18. A point charge $-q$ is placed inside cavity of a metallic sphere of radius R . Radius of the cavity is r . Potential at point P will be -



- (A) $\frac{-q}{4\pi\epsilon_0 y}$
 (B) $\frac{-q}{4\pi\epsilon_0 y} + \frac{q}{4\pi\epsilon_0 x}$
 (C) $\frac{-q}{4\pi\epsilon_0 x}$
 (D) $\frac{q}{4\pi\epsilon_0 y} - \frac{q}{4\pi\epsilon_0 x}$
19. A particle is projected from bottom of a blind narrow tunnel of depth $\frac{R}{4}$ made along a radius inside the earth. If minimum velocity needed to escape out of the gravitational field of the earth is $\frac{1}{4}\sqrt{\frac{xGM}{R}}$. Find x. M and R are mass and radius of the earth respectively.

- (A) 39
 (B) 78
 (C) 86
 (D) 0

20. A ray of light incidents at an angle of incidence θ in water at water-air interface. Graph between angle of deviation (δ) and angle of incidence (θ) will be-



SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

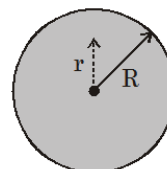
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

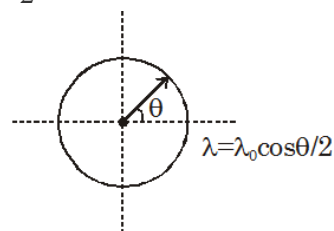
Negative Marks : -1 If wrong option is selected.

1. A body is thrown from surface of earth with velocity $\left(\sqrt{\frac{gR_e}{2}}\right)$ [R_e radius of earth] at some angle from vertical. Maximum height reach by the body is $\frac{R_e}{4}$. If angle of projection is $\cos^{-1}\left[\frac{\sqrt{x}}{4}\right]$ find the value of x.

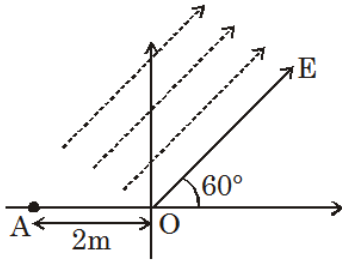
2. A uniform solid sphere having mass density is ρ . The ratio of gravitational pressure at a distance $r = R/2$ to the centre of sphere is $\frac{3x}{8}$, the value of x will be



3. Linear charge density of ring of radius 'R' given as $\lambda = \lambda_0 \cos \frac{\theta}{2}$. The potential at centre of ring will be

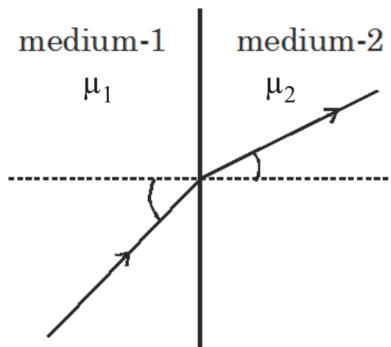


4. Uniform $\vec{E} = 100\text{V/m}$ is directed as shown in figure. The potential difference between $V_A - V_O$ is

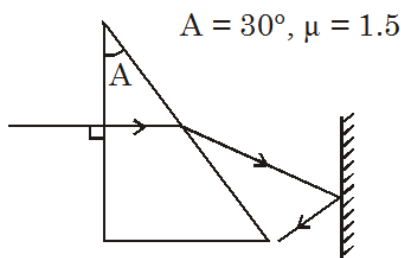


5. A ray incident on medium boundary as shown in figure. Unit vector in direction of incident ray is given as $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ & unit vector of ray in medium (2) is given by $\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$. The

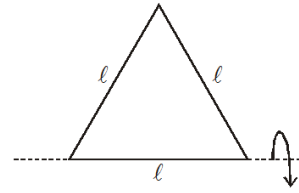
ratio of $\frac{\left(\frac{\mu_1}{\mu_2}\right)^2 + \frac{1}{2}}{\left(\frac{\mu_1}{\mu_2}\right)^2 - \frac{1}{2}}$ is



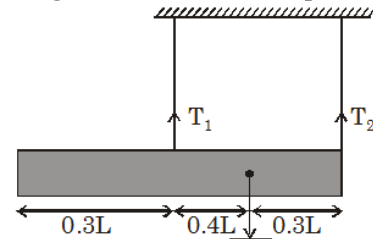
6. A ray incident normal to the surface of prism as shown in figure. Angle of deviation of ray from the mirror w.r.t incident ray is $210^\circ - \sin^{-1} \frac{x}{4}$, find the value of x.



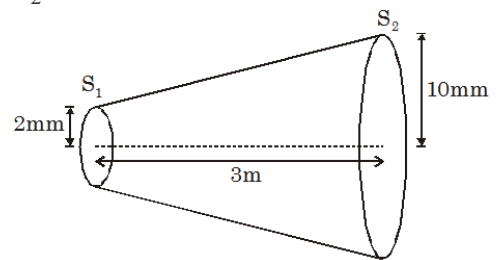
7. Three identical rod each of mass m & length ℓ form a equilateral triangle. Moment of inertia about given axis is $\frac{m\ell^2}{x}$, value of x will be :-



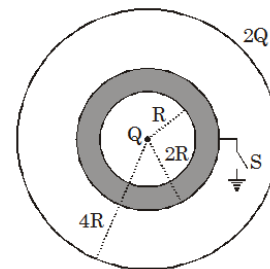
8. In given diagram a uniform plank weight is 500 N. Find out weight W if T_1 & T_2 are equal.



9. Figure shows segment of a current carrying wire made of copper in steady state. If ratio of magnitudes of electric field at cross-section S_1 and S_2 is $5n$. Find value of n .



10. A thick conducting shell of inner radius R and outer radius $2R$ has charge Q at its center. It is covered by a thin conducting shell of radius $4R$ which has charge $2Q$. If charge flown into the earth when switch is closed is nQ , find n .



PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

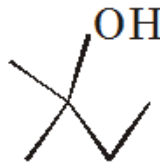
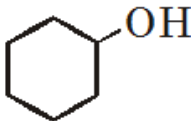
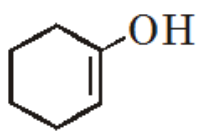
Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

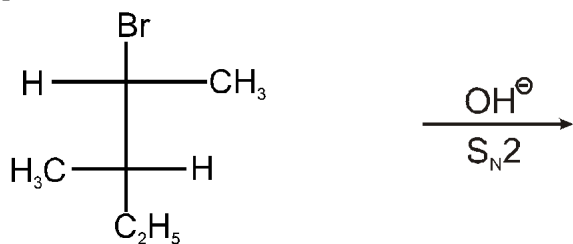
Negative Marks : -1 If wrong option is selected.

- The correct order of second I.E. of C, N, O and F are in the order
(A) $F > O > N > C$
(B) $C > N > O > F$
(C) $O > N > F > C$
(D) $O > F > N > C$
- Calamine malachite, magnetite and cryolite, respectively are:
(A) $ZnSO_4$, $CuCO_3$, Fe_2O_3 , AlF_3
(B) $ZnSO_4$, $Cu(OH)_2$, Fe_3O_4 , Na_3AlF_6
(C) $ZnCO_3$, $CuCO_3 \cdot Cu(OH)_2$, Fe_3O_4 , Na_3AlF_6
(D) $ZnCO_3$, $CuCO_3$, Fe_2O_3 , Na_3AlF_6
- Select the incorrect IUPAC name from the following
(A) $H[AuCl_4]$: Hydrogen tetrachloridoaurate(III)
(B) $[PtCl_6]^{2-}$: Hexachloridoplatinate(IV) ion
(C) $[Cu(NH_3)_4]SO_4$: Tetraamminecopper(II) sulphate
(D) $[Cr(en)_3]Cl_3$: Tris (ethylenediamine)chromium(III) chloride

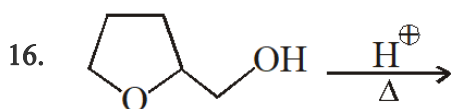
- Spin only magnetic moment in BM of $[Fe(CO)_4(C_2O_4)]^+$ is :
(A) 5.92 (B) 0
(C) 2.82 (D) 1.73
- Which of the following complex is optically active :
(A) $trans-[Co(NH_3)_4Cl_2]^+$
(B) $[Cr(H_2O)_6]^{3+}$
(C) $cis-[Co(NH_3)_2(en)_2]^{3+}$
(D) $trans-[Co(NH_3)_2(en)_2]^{3+}$
- Select **INCORRECT** order against property indicated—
(A) $AsCl_3 < SbCl_3 < BiCl_3$ (relative stability)
(B) $ZnO > MgO > CaO > BaO$ (Covalent character)
(C) $NaF < MgF_2 < AlF_3$ (M.P.) Melting point
(D) $BeF_2 > MgF_2 > CaF_2 > SrF_2 > BaF_2$ (water solubility)
- At 300 K, the vapour pressure of an ideal solution containing 3 mole of A and 2 mole of B is 600 torr. At the same temperature, if 1.5 mole of A & 0.5 mole of C (non-volatile) are added to this solution the vapour pressure of solution increases by 30 torr. What is the value of P_B^0 ?
(A) 940 torr
(B) 405 torr
(C) 90 torr
(D) 100 torr

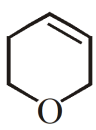
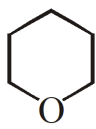
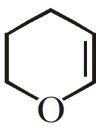
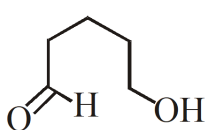
8. What is the correct sequence of osmotic pressures of 0.01 M aq. solutions of :
 (i) $\text{Al}_2(\text{SO}_4)_3$ (π_1)
 (ii) Na_3PO_4 (π_2)
 (iii) BaCl_2 (π_3)
 (iv) Glucose (π_4)
 [Assume 100% ionisation of electrolytes]
 (A) $\pi_4 > \pi_2 > \pi_3 > \pi_1$
 (B) $\pi_3 > \pi_4 > \pi_2 > \pi_1$
 (C) $\pi_3 > \pi_4 > \pi_1 > \pi_2$
 (D) $\pi_1 > \pi_2 > \pi_3 > \pi_4$
9. Find ratio of mole fractions of gases N_2 & O_2 dissolved in water from air at 25° & 1 atm. Assume air is 20 % by volume of O_2 & 80% by volume of N_2
Given : $K_H(\text{N}_2) = 2 \times 10^4 \text{ atm}$
 $K_H(\text{O}_2) = 10^4 \text{ atm}$
 (A) 8 : 1 (B) 1 : 8
 (C) 2 : 1 (D) 1 : 1
10. For a process $\text{A} + \text{B} \rightarrow \text{P}$, the reaction is second order with respect to A and zero order with respect to B. When 1 mole each of A & B are taken in 1 litre vessel the initial rate is $1 \times 10^{-2} \text{ mol/L-sec}$. The rate of reaction when 50% of the reactant have been converted to product would be -
 (A) $1 \times 10^{-2} \text{ mol/L-sec}$
 (B) $2.5 \times 10^{-3} \text{ mol/L-sec}$
 (C) $5 \times 10^{-2} \text{ mol/L-sec}$
 (D) $5 \times 10^{-3} \text{ mol/L-sec}$
11. A vessel contains He gas at 20 atm and CH_4 gas at 'P' atm. When a pin-hole is made in the vessel, the gases coming out initially contains 80% He gas, by mole. The value of 'P' is :-
 (A) 1 atm (B) 40 atm
 (C) 2.5 atm (D) 10 atm
12. A real gas behaves like an ideal gas at 132°C in low pressure range. Critical temperature for this gas is -
 (A) 120 K (B) 39 K
 (C) 132 K (D) 15 K
13. In a CCP lattice of X & Y, X atoms are present at the corners while Y atoms are at face centers. Then the formula of the compound would be if one of the X-atoms from a corner in each unit cell is replaced by Z-atom :
 (A) $\text{X}_7 \text{Y}_{24} \text{Z}_2$ (B) $\text{X}_7 \text{Y}_{24} \text{Z}$
 (C) $\text{X}_{24} \text{Y}_7 \text{Z}$ (D) $\text{X} \text{Y}_{24} \text{Z}$
14. Which of the following alcohol is expected to dehydrate using conc. H_2SO_4 at fastest rate ?
 (A) 
 (B) 
 (C) 
 (D) 

15. In the following reaction the most probable product will be :

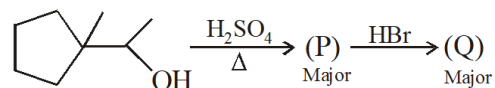


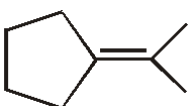
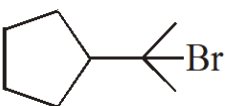
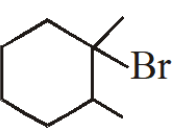
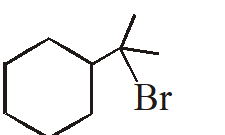
- (A) $\begin{array}{c} \text{OH} \\ | \\ \text{H} - \text{C} - \text{CH}_3 \\ | \\ \text{H}_3\text{C} - \text{C} - \text{H} \\ | \\ \text{C}_2\text{H}_5 \end{array}$ (B) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H}_3\text{C} - \text{C} - \text{H} \\ | \\ \text{C}_2\text{H}_5 \end{array}$
- (C) $\begin{array}{c} \text{C}_2\text{H}_5 \\ | \\ \text{H} - \text{C} - \text{CH}_3 \\ | \\ \text{H}_3\text{C} - \text{C} - \text{H} \\ | \\ \text{OH} \end{array}$ (D) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H}_3\text{C} - \text{C} - \text{H} \\ | \\ \text{C}_2\text{H}_5 \end{array}$

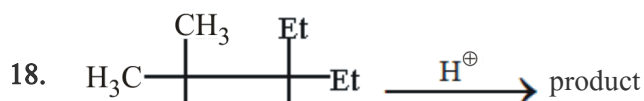


- (A) 
- (B) 
- (C) 
- (D) 

17. The final major product (Q) of the given sequence of reaction :

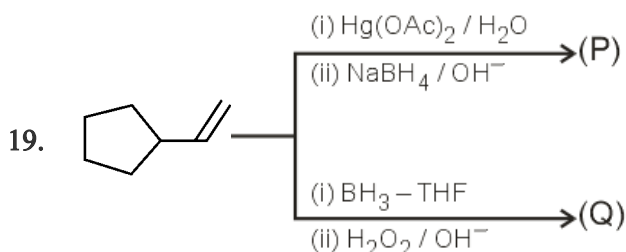


- (A)  (B) 
- (C)  (D) 



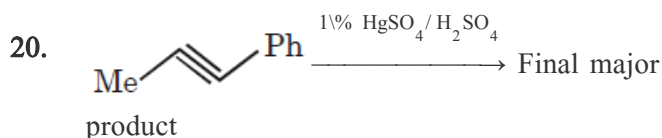
Major product of the reaction is :

- (A) $\text{CH}_3 - \text{C}(=\text{O}) - \text{C}(\text{Et})(\text{CH}_3) - \text{Et}$
- (B) $\text{CH}_3 - \text{C}(\text{CH}_3)(\text{Et}) - \text{C}(=\text{O}) - \text{Et}$
- (C) $\text{CH}_3 - \text{C}(\text{CH}_3)(\text{CH}_3) - \text{C}(=\text{O}) - \text{CH}_3$
- (D) $\text{Et} - \text{C}(\text{Et})(\text{Et}) - \text{C}(=\text{O}) - \text{Et}$



Choose the correct statement ?

- (A) (P) & (Q) are chain isomers
(B) (P) will give immediate turbidity with Lucas reagent
(C) (P) & (Q) are identical
(D) (P) & (Q) are positional isomers



Select correct statement for above reaction.

- (A) Final compound show tautomerism
(B) Final compound show geometrical isomerism
(C) Final compound show optical isomerism
(D) Final compound not show structural isomerism

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Find out the no. of correct statements :

- (a) Bond length $N_2^+ >$ Bond length N_2
(b) Bond length $NO^+ <$ Bond length of NO
(c) Bond length $CN^- <$ Bond length of CN
(d) Bond length $O_2^- <$ Bond length of O_2^{-2}
(e) Bond length $O_2 >$ Bond length of O_2^+
(f) Bond length $B_2 >$ Bond length of B_2^-

2. The number of $2c - 2e$ bonds present in a dimer molecule of $Al(CH_3)_3$ is-

3. Find the total number of type of isomerism (from given below) shown by the complex $[Fe(NH_3)_4(H_2O)Br](NO_3)_2$

Ionization isomerism, linkage isomerism, ligand isomerism, Hydrate isomerism, geometrical isomerism

4. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is:

5. For a reaction $A \rightarrow P$

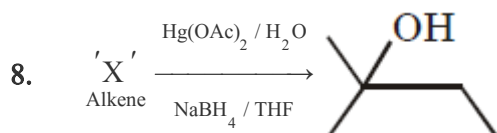
Half life measured for two different values of initial concentrations of A, $5 \times 10^{-3} M$ and $25 \times 10^{-4} M$ are 1 and 8 hrs respectively. If initial concentration is adjusted to $1.25 \times 10^{-3} M$, what is the half-life? (in hrs)

6. In the following reaction : $x A \rightarrow y B$, following equation is given :

$$\log \left[-\frac{d[A]}{dt} \right] = \log \left[\frac{d[B]}{dt} \right] + 0.3,$$

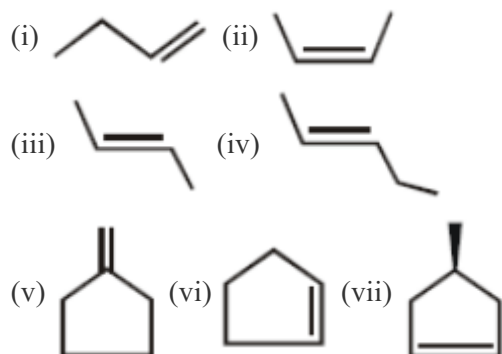
Find ratio $\left(\frac{y}{x} \right)$: [Given $\log 2 = 0.3$]

7. A metal 'A' exists as BCC structure having uncovered length between the adjacent atoms along the edge in a unit cell equal to $1.2 \text{ \AA}$. If molar mass of the metal is 307.2 gm/mole then calculate density of the crystal in gm/ml .
[Use : $N_A = 6 \times 10^{23}$ and $\sqrt{3} = 1.7$]

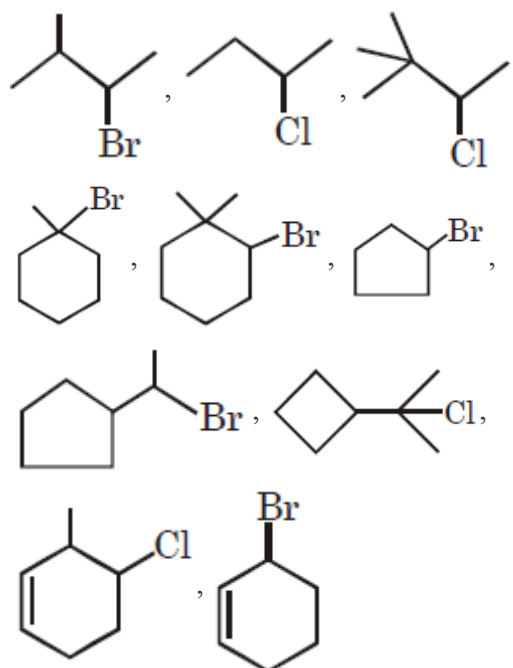


Identify possible number of alkenes (X) which can give, Given product by oxymercuration demercuration :

9. How many of the following alkenes will form racemic mixture when reacts with $\text{Br}_2 / \text{CCl}_4$:



10. How many substrate(s) will show significant rearrangement during $\text{S}_{\text{N}}1$ reaction :



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. If the angle between the pair of straight lines represented by the equation

$$x^2 - 3xy + \lambda y^2 + 3x - 5y + 2 = 0 \text{ is } \tan^{-1} \left(\frac{1}{3} \right),$$

where λ is a non-negative real number. Then λ is-

- (A) 2 (B) 0 (C) 3 (D) 1

2. Let $g(x) = x^4 + \frac{1}{x^4}$ & $f : [2, \infty) \rightarrow \mathbb{R}$ such that

$$f(g(x)) = x^{12} + \frac{1}{x^{12}} \text{ then } f'(x) \text{ is}$$

(A) $3x^2 + x + 1$

(B) $3x^2 - 3$

(C) $x^5 + \frac{1}{x^5}$

(D) $x^{10} - x^3$

3. Let $F : (1, \infty) \rightarrow \mathbb{R}$ be a function defined as

$$F(x) = \int_x^{x^2} \frac{dt}{\ln t}. \text{ Which of the following}$$

statements are **INCORRECT**?

(A) $F(x)$ is strictly decreasing

(B) $F(x)$ is strictly increasing

(C) $F'(e) = e - 1$

(D) $F(x)$ is a one - one function

4. The value of the definite integral

$$\int_{\pi/24}^{5\pi/24} \frac{dx}{1 + \sqrt[3]{\tan 2x}} \text{ is :}$$

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{12}$

(D) $\frac{\pi}{18}$

5. $\int_0^{\frac{\pi^2}{4}} \left(\frac{2}{\operatorname{cosec} \sqrt{x}} + \frac{\sqrt{x}}{\sec \sqrt{x}} \right) dx$ is equal to

(A) $\frac{\pi^2}{4}$

(B) $\frac{\pi^2}{2}$

(C) π^2

(D) $\frac{\pi}{8}$

6. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin(\sqrt{t}) dt}{x^3}$ is equal to

(A) $\frac{1}{3}$

(B) 1

(C) $\frac{2}{3}$

(D) does not exist

7. The number of solution(s) of the equation

$$\frac{d}{dx} \int_{\cos x}^{\sin x} \frac{dt}{1-t^2} = 2\sqrt{2} \text{ in } [0, \pi] \text{ is :}$$

(A) 4

(B) 3

(C) 2

(D) 0

8. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{(2\tan^{-1}e^x - \frac{\pi}{2})}{1+x^2} dx$ is equal to -
- (A) $\frac{\pi}{2}$
 (B) $\frac{\pi^2}{4}$
 (C) $e^{\frac{\pi^2}{2}}$
 (D) 0
9. The integral $\int \frac{\sqrt[3]{(1+2x^3)^2}}{x^6} dx$ is (Where 'C' is an arbitrary constant)
- (A) $-\frac{1}{5} \left(1 + \frac{2}{x^3}\right)^{5/3} + C$
 (B) $\frac{1}{5} \left(2 + \frac{1}{x^3}\right)^{5/3} + C$
 (C) $-\frac{1}{5} \left(2 + \frac{1}{x^3}\right)^{5/3} + C$
 (D) $\frac{1}{5} \left(1 + \frac{2}{x^3}\right)^{5/3} + C$
10. $\int \frac{x^5 - 2x^6}{(x^{15} - x^{17} + x^{16})^{1/3}} dx$, is (where 'C' is any arbitrary constant)
- (A) $\frac{3}{2}(1 - x^2 + x)^{2/3} + C$
 (B) $\frac{2}{3}(1 - x^2 + x)^{2/3} + C$
 (C) $-\frac{3}{2}(1 - x^2 + x)^{2/3} + C$
 (D) $\frac{3}{2}(1 - x^2 + x)^{-2/3} + C$
11. If $\int \frac{1}{x+x^5} dx = f(x) + c$, then the value of $\int \frac{x^4}{x+x^5} dx$ is :
- (A) $\log x - f(x) + c$
 (B) $f(x) + \log x + c$
 (C) $f(x) - \log x + c$
 (D) none of these
12. The function $f(x) = x(x+4)e^{-x/2}$ has its local maxima at $x = a$. Then
- (A) $a = -4$
 (B) $a = 2\sqrt{2}$
 (C) $a = 1 - \sqrt{3}$
 (D) $a = -1 + \sqrt{3}$
13. Let $f(x) = \begin{cases} \frac{ax^3+bx^2+cx+d}{x} & \forall x \neq 0 \\ 2 & x = 0 \end{cases}$ is a continuous function where a, b, c, d are in Arithmetic progression. If $x = p$ be the point of local minimum of $f(x)$ then identify the **INCORRECT** option ? (where $[.]$ represents greatest integer function and $\{.\}$ represents fractional part function)
- (A) $[p] = -1$
 (B) $\{p\} = \frac{2}{3}$
 (C) minimum value of $f(x)$ is $\frac{4}{3}$
 (D) $\{p\} = \frac{1}{3}$

14. A polynomial $f(x)$ of the least degree, has a local maximum value equal to 6 at $x = 1$, and a local minimum value equal to 2 at $x = 3$, then identify the **INCORRECT** option.

(where $[.]$ represent greatest integer function)

- (A) The equation $f(x) = 3$, has three distinct roots.
 (B) $f(x)$ is strictly decreasing in $1 < x < 2$
 (C) $f(0) = 1$
 (D) If α is real root of $f(x) = 0$, then $[\alpha] = -1$

15. The maximum slope of the curve

$$y = -x^3 + 3x^2 + 9x - 27 \text{ is :-}$$

- (A) 0
 (B) 12
 (C) 16
 (D) 32
16. A thrice differentiable function $y = f(x)$ is such that its graph cuts the curve $y = ax^2 + bx + c$ at 10 distinct points. The minimum number of distinct roots of $f'''(x) = 0$ is :
- (A) 0
 (B) 1
 (C) 8
 (D) 7

17. Let $P(x) = x^2 + bx + c$ be a quadratic polynomial with real coefficients such that $\int_0^1 P(x) dx = 1$

and $P(x)$ leaves remainder 5 when it is divided by $(x - 2)$. Then the value of $9(b + c)$ is equal to

- (A) 9 (B) 15
 (C) 7 (D) 11

18. The value of integral $\int_0^1 x(1-x)^{2022} dx$ is

- (A) $\frac{1}{(2022)(2023)}$
 (B) $\frac{2022}{2023}$
 (C) $\frac{1}{(2023)(2024)}$
 (D) $\frac{2023}{2024}$

19. Tangent from $\left(0, -\frac{1}{2}\right)$ to $y^2 = x^3$ passes through -

- (A) (4, 8)
 (B) (1, -1)
 (C) (1, 0)
 (D) (3, 4)

20. If $f(x) = 2x^3 + 7x - 5$ then $f^{-1}(4)$ is :

- (A) equal to 1
 (B) equal to 2
 (C) equal to $1/3$
 (D) non existent

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct integer value (in decimal notation, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to :

2. The value of $I = \int_0^1 \left[\frac{\sin x}{x} \right] dx$ (where $[.]$ represents greatest integer function)

3. If $I = \int_0^1 e^x (x-1)^3 dx$, then absolute value of $[I]$ (where $[.]$ denotes greatest integer function), is

4. If $I = \int_0^{7\pi + \frac{2\pi}{3}} |\cos x| dx$, then $[I]$ is equal to (where $[.]$ denotes greatest integer function):

5. The value of $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sec^3(t^2) \cdot dt}{x \sin x}$ is

6. Consider $I_1 = \int_0^2 \sqrt{x^3 + 1} dx$ &

$I_2 = \int_1^3 \sqrt[3]{x^2 - 1} dx$, then $I_1 + I_2$ is equal to

7. If $\int e^x \left(\frac{x+2}{x+4} \right)^2 dx = e^x \left(\frac{x-\alpha}{x+\beta} \right) + C$ (where C is integration constant), then $|\beta - \alpha|$ is equal to (where α, β are constant)

8. If α, β, γ are roots of $x^3 + x^2 - 5x - 1 = 0$, then value of $[\alpha] + [\beta] + [\gamma]$ is (where $[.]$ denotes greatest integer function)

9. Let $f(x) = (x-1)^{100} (x-2)^{2(99)} (x-3)^{3(98)} \dots (x-100)^{100}$. If $P = \frac{f'(101)}{f(101)}$, then P equals

10. If $\int \frac{xe^x + e^x}{(1 + xe^x)^2} dx = f(x) + C$, where C is a constant independent of x and $f(0) = 1$, then value of $\lim_{x \rightarrow \infty} f(x)$ is

Space for Rough Work

